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A case study on

Smart Restaurant Ordering Management System

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2. Executive Summary

The Smart Restaurant Ordering Management System is a database-driven project developed using Microsoft SQL Server to automate and manage restaurant operations efficiently. The system handles dining management, menu categorization, customer orders, and payment processing. This project replaces the traditional manual ordering system with a structured, secure, and optimized database solution using tables, views, stored procedures, functions, and triggers to ensure data integrity and smooth transaction processing.

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3. Introduction

Modern restaurants require fast and accurate order management systems to improve customer satisfaction and operational efficiency. Manual systems often result in errors, delayed service, and poor record management. This project focuses on designing and implementing a relational database system that automates restaurant ordering, customer management, and payment tracking.

The system is built using SQL Server and follows proper database normalization, constraints, and business logic implementation.

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4. Problem Statement

Traditional restaurant management systems face several challenges such as:

- Manual order taking and billing errors
- Difficulty in tracking customer orders
- Inefficient payment management
- Lack of data security and consistency

These issues lead to operational inefficiency, customer dissatisfaction, and poor reporting.

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5. Objectives of the Project

The main objectives of this project are:

- To design a structured restaurant ordering database system

- To manage dining tables and menu categories efficiently
- To automate order processing and payment handling
- To ensure data integrity using constraints and triggers
- To improve performance using views, functions, and stored procedures

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6. Scope of the Project

Included:

- Dining table management
- Menu category and menu item management
- Customer order processing
- Payment method and payment tracking
- Database-level business logic implementation

Excluded:

- Front-end or mobile application
- Online delivery integration

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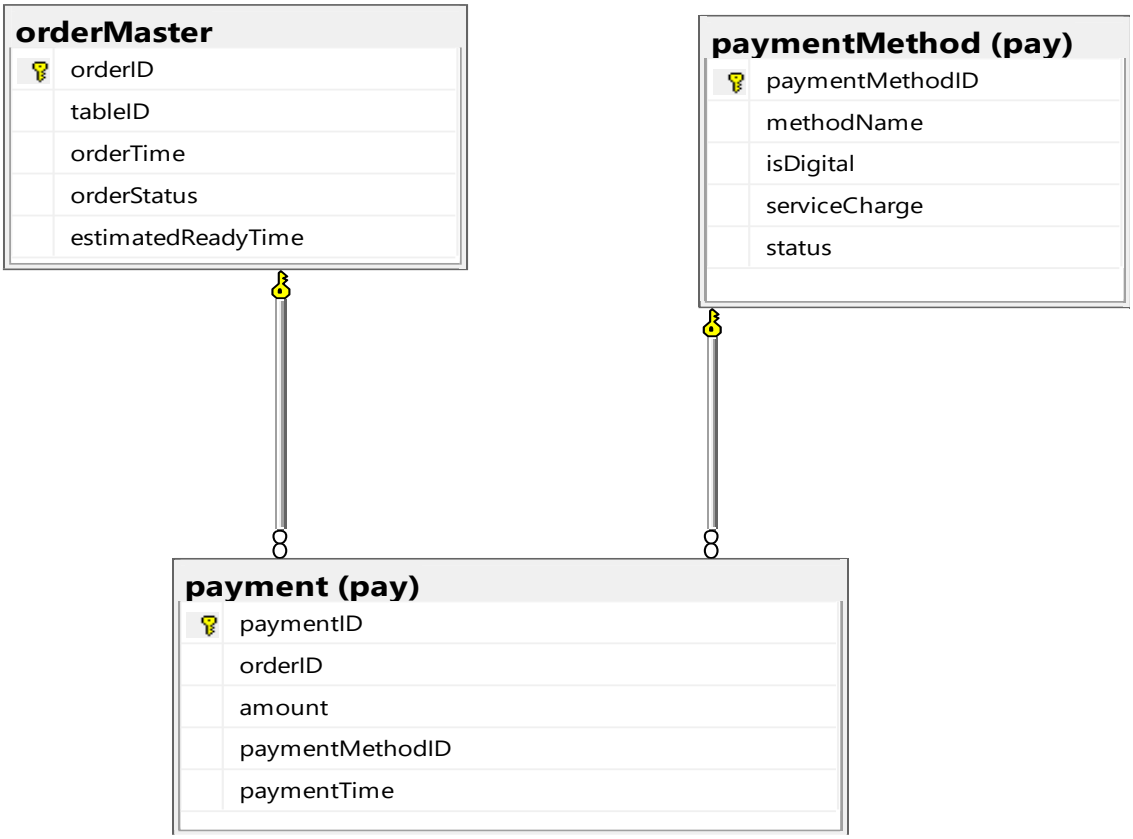
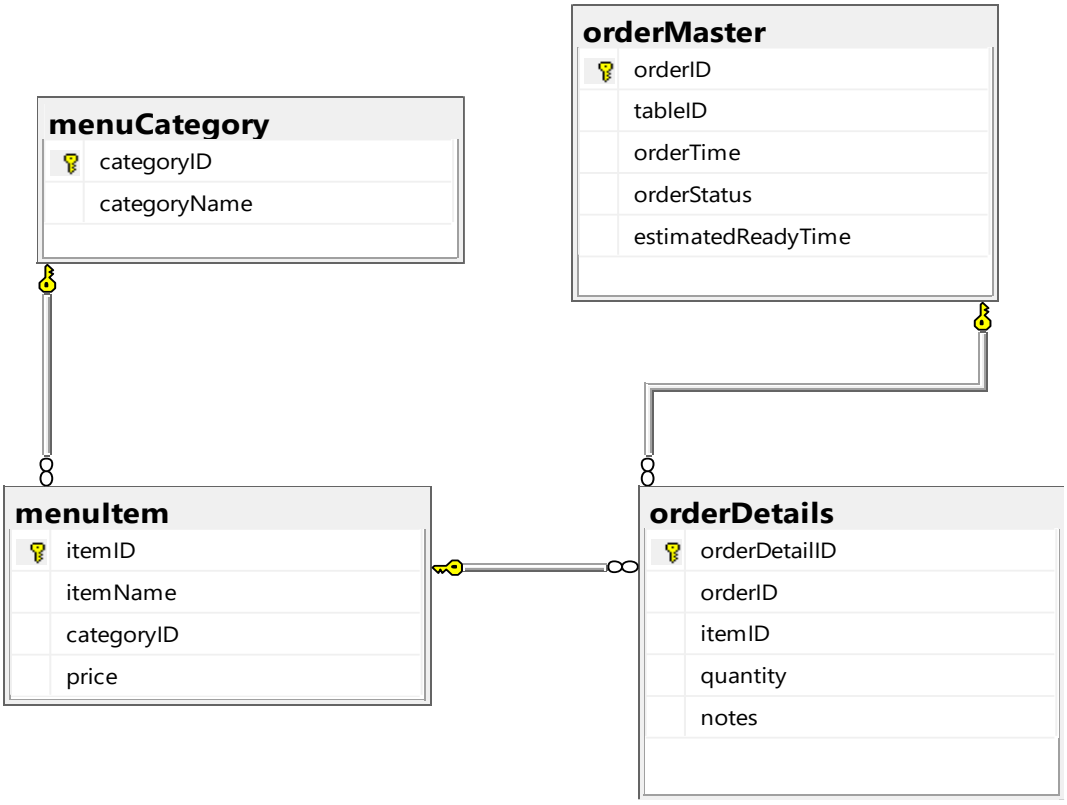
7. Database Design

Tables Used

1. Restaurant Table
2. Menu Category
3. Menu Item
4. Order Master
5. Order Details
6. Customers
7. Payment Method
8. Payment

The database follows relational integrity using **Primary Keys**, **Foreign Keys**, and **constraints** to maintain consistency.


orderMaster
orderID
tableID
orderTime
orderStatus
estimatedReadyTime
paymentMethod (pay)
paymentMethodID
methodName
isDigital
serviceCharge
status
payment (pay)
paymentID
orderID
amount
paymentMethodID
paymentTime


orderMaster
orderID
tableID
orderTime
orderStatus
estimatedReadyTime
menuCategory
categoryID
categoryName
menuItem
itemID
itemName
categoryID
price
orderDetails
orderDetailID
orderID
itemID
quantity
notes

8. Methodology

The project was developed following these steps:

1. Requirement analysis
2. Database design and normalization
3. Table creation using SQL Server
4. Implementation of constraints and relationships
5. Development of views for reporting
6. Creation of user-defined functions
7. Implementation of stored procedures for CRUD operations
8. Use of triggers for automatic actions and validation

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9. Features Implemented

9.1. Data Definition Language (DDL)

SECTION 01: CREATE OBJECTS

- Create Database
- Create Sequence
- Create Tables with Constraints
- Create Views, Stored Procedures, Functions, Triggers

SECTION 02: ALTER OBJECTS

SECTION 03: DROP OBJECTS

9.2. Data Manipulation Language (DML)

SECTION 01: INSERT DATA

SECTION 02: USING SEQUENCES

SECTION 03: INDEX

SECTION 04: INSERT/UPDATE/DELETE USING VIEWS

SECTION 05: STORED PROCEDURES

SECTION 06: RETRIEVE DATA

SECTION 07: TRIGGERS

SECTION 08: QUERIES

9.3. Testing and Deployment

1. **Test DDL and DML**
   - Test for data integrity and performance.
   - Use transactions and error handling (TRY-CATCH) for robust execution.
2. **Simulate Real Scenarios**
   - Event registrations, donations, and reporting.

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10. Implementation

The database was implemented using Microsoft SQL Server. All tables were created with proper relationships. Stored procedures and functions were used to reduce redundancy and improve performance. Triggers were implemented to automate critical operations and ensure data consistency across related tables.

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11. Results and Analysis

After implementation:

- Order processing became faster and more accurate
- Data redundancy was minimized
- Payment records became more reliable
- Reporting and analysis became easier using views

The system successfully achieved its objectives.

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12. Limitations

- No graphical user interface (GUI)

- Manual data input required
- No real-time mobile or web integration

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13. Conclusion

The Smart Restaurant Ordering Management System successfully demonstrates how SQL Server can be used to design a reliable and efficient restaurant database system. By using tables, views, stored procedures, functions, and triggers, the system ensures data integrity, improves performance, and reduces manual errors. This project provides a strong foundation for future expansion into a full-scale restaurant management application.

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14. Recommendations

- Develop a web or mobile front-end
- Add role-based user access
- Integrate online payment gateways
- Implement real-time order tracking